Least-squares Parameter Estimation (Åström & Wittenmark, Example 2.1)

Adaptive Control, Homework 1

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology • Autumn 1404 (2025)

English summary of my original Persian report, which is kept as a PDF in this folder.

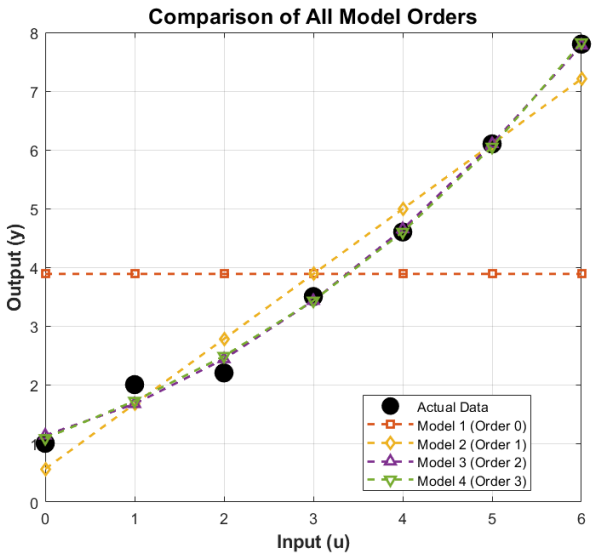
Problem

Seven input–output pairs, $u = 0, \ldots, 6$ and $y = (1,\ 2,\ 2.2,\ 3.5,\ 4.6,\ 6.1,\ 7.8)$, are fitted with polynomial models of order 0 to 3, $y = b_0 + b_1 u + \cdots + b_3 u^3$, by least squares: $\hat{\theta} = (\Phi^T \Phi)^{-1} \Phi^T y$.

Results (`main.m`, re-run in R2023b)

The loss is $V = \frac{1}{2} \sum_i (y_i - \hat{y}_i)^2$.

Model	b_0	b_1	b_2	b_3	V
Order 0	3.8857				17.8043
Order 1	0.5643	1.1071			0.6436
Order 2	1.1357	0.4214	0.1143		0.0950
Order 3	1.0857	0.5881	0.0393	0.0083	0.0875



Conclusion

The loss drops sharply from order 0 to order 2 and only slightly after that. Order 1 already captures the trend, and order 2 fits almost exactly. Going to order 3 adds a parameter for very little gain, so model order has to be traded off against the risk of over-fitting.